ENT ATLAS

IMAGING

Imaging conditions: Larynx and neck

Key facts and case images for the core conditions, then the advanced and rare

· November 2026

IMAGING

Core conditions

The common conditions every resident should recognise, each with its key imaging findings, top differentials, a short checklist and representative scans.

NECK: TRIANGLES, FASCIA & NODES · INFECTION & INFLAMMATION

Abscess of the Neck

IMAGING FINDINGS

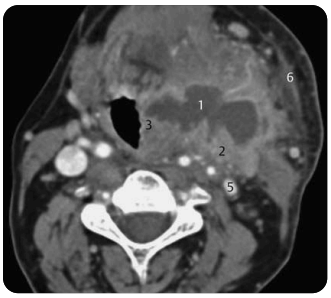
- Low-density collection with a well-formed enhancing rim
- Surrounding cellulitis and fat stranding
- The source: tooth, tonsil, node, stone or foreign body

TOP DIFFERENTIALS

- Phlegmon (no fluid centre)
- Necrotic metastatic node
- Infected congenital cyst

CHECKLIST

- Airway, mediastinum and jugular vein
- Which space: decides the drainage route



CT with intravenous contrast enhancement, axial.A large abscess (1) with a thick, contrast-enhanced capsule (2). The abscess is extendin…

NECK: TRIANGLES, FASCIA & NODES · INFECTION & INFLAMMATION

Parapharyngeal Abscess

IMAGING FINDINGS

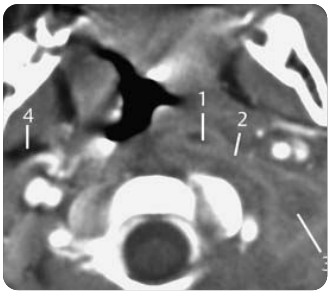
- Collection in the parapharyngeal fat, usually from the tonsil
- Medial bulge of the pharyngeal wall
- Trismus from medial pterygoid involvement

TOP DIFFERENTIALS

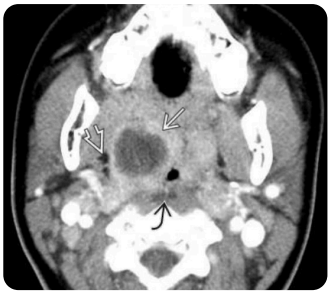
- Peritonsillar abscess
- Retropharyngeal abscess
- Deep lobe parotid infection

CHECKLIST

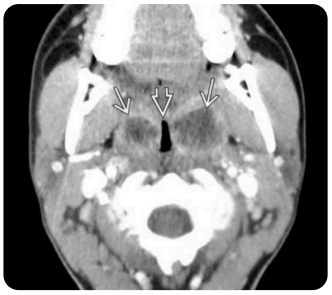
- Carotid space and jugular vein (Lemierre)
- Spread to the mediastinum



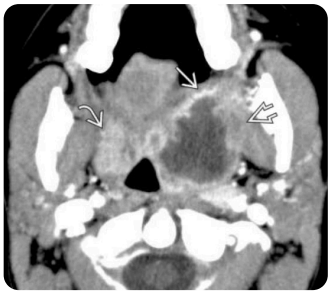
CT with intravenous contrast enhancement, axial. Note the bulging left pharyngeal wall, as a result of a parapharyngeal abscess (1) which is…



Axial CECT shows a large, low-density tonsillar abscess. No extension through the capsule into PPS is present. The left tonsil is prominent…



Axial CECT reveals bilateral tonsillar abscesses, the left larger than the right. Note the mass effect on both PPS but no extension of inflammatory…



Axial CECT demonstrates left tonsillar and peritonsillar abscess with extension through the capsule into the posterior buccal space and medial…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 28.

NECK: TRIANGLES, FASCIA & NODES · INFECTION & INFLAMMATION

Cervical Lymphadenitis

IMAGING FINDINGS

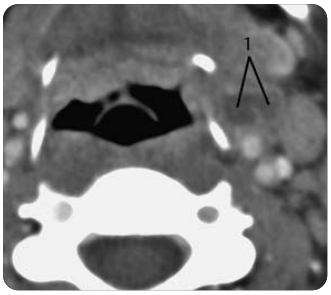
- Enlarged nodes with preserved fatty hilum (reactive)
- Suppurative node: thick rim around a fluid centre with surrounding inflammation
- Matted necrotic nodes in tuberculosis

TOP DIFFERENTIALS

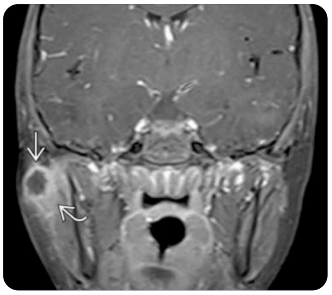
- Metastatic squamous carcinoma
- Second branchial cleft cyst
- Atypical mycobacteria

CHECKLIST

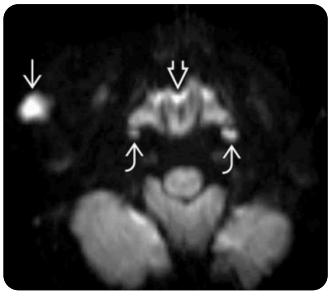
- Ultrasound first in children
- Progression to abscess



CT with intravenous contrast enhancement, axial. Various enlarged level III lymph nodes are present near the internal carotid artery and jugular…



Coronal T1 C+ FS MR in an 8-year-old boy with right cheek swelling shows a peripherally enhancing, centrally necrotic intraparotid mass with…



Axial DWI trace in the same patient shows the parotid mass to have central reduced diffusivity, typical of suppurative adenitis. Note reactive…



Axial CECT demonstrates a well-defined, low-density submandibular mass with thin peripheral rim enhancement. Suppurative node is…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 314.

NECK: TRIANGLES, FASCIA & NODES · TUMOURS

Lymphoma, Non-Hodgkin

IMAGING FINDINGS

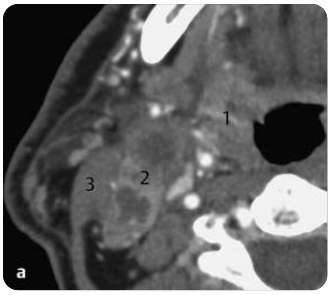
- Large, homogeneous, non-necrotic nodes in many chains
- Waldeyer ring involvement
- Strong diffusion restriction

TOP DIFFERENTIALS

- Reactive nodes
- Tuberculosis
- Metastatic carcinoma

CHECKLIST

- Extranodal sites: tonsils, sinuses, orbit, thyroid
- Core or excision biopsy, not FNA alone



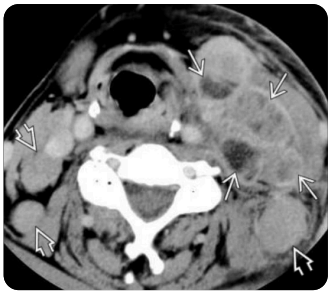
CT with intravenous contrast enhancement, axial. On this slice, an area of different structure with enhancement is seen at the base of…



Axial CECT shows multiple large, round, solid lymph nodes in every chain of the suprahyoid neck. Bilateral nodes are evident in levels IIA, IIB, and…



Axial CECT in a different patient shows nodal and tonsillar lymphoma. Level II nodal mass is partially necrotic, with extranodal infiltration of fat and…



Axial CECT demonstrates massive left nodal conglomerate, some with necrosis. Surrounding induration and stranding of fat in left neck suggests…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 336.

NECK VESSELS & NERVES · TUMOURS

Carotid Body Paraganglioma

IMAGING FINDINGS

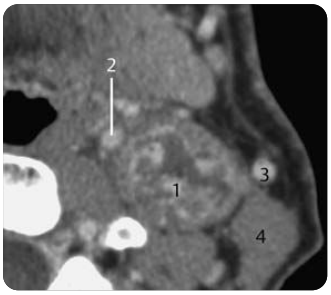
- Avidly enhancing mass in the carotid bifurcation
- Splaying of the internal and external carotid arteries
- Flow voids ("salt and pepper") on MRI

TOP DIFFERENTIALS

- Vagal paraganglioma (pushes ICA forwards)
- Carotid space schwannoma
- Jugulodigastric node; pseudoaneurysm

CHECKLIST

- Shamblin class: degree of carotid encasement
- Look for multiple paragangliomas



Patient with a slowly progressive lesion in the neck.

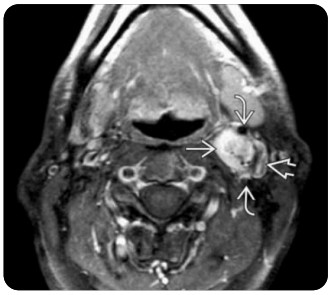
Dunnebier, Imaging for Otolaryngologists, Fig. 9.10 a



Axial CECT shows a classic carotid body paraganglioma with avid, fairly uniform enhancement. Notice the clear definition of the tumor sitting in the…



Axial T1WI MR shows a rounded mass in the left carotid space, between 2 flow voids representing the ICA and ECA. Small internal foci of signal void…



Axial T1WI C+ FS MR in the same patient shows intense enhancement of a solid mass between the splayed branches of the carotid artery. The…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 140.

NECK: TRIANGLES, FASCIA & NODES · CONGENITAL

Branchial Cleft Cyst

IMAGING FINDINGS

- Cyst at the angle of the mandible in front of the sternocleidomastoid
- Lateral to the carotid sheath, behind the submandibular gland
- Thick wall when infected

TOP DIFFERENTIALS

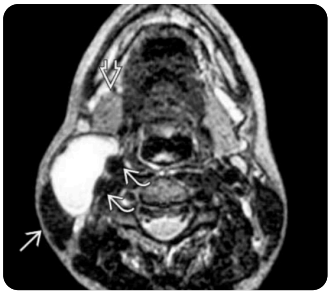
- Cystic nodal metastasis in an adult
- Lymphatic malformation
- Thymic cyst

CHECKLIST

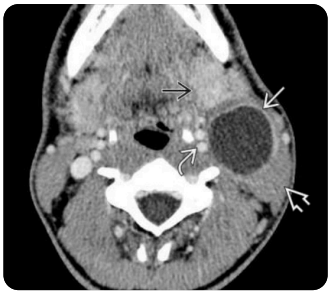
- In adults: exclude HPV oropharyngeal cancer
- Tract towards the tonsillar fossa



CT with intravenous contrast enhancement, coronal (a) and axial (b).
Dunnebier, Imaging for…



Axial T2WI MR shows a well-defined hyperintense mass anterior to the right SCM, posterolateral to the right submandibular gland, and…



Axial CECT in a 17-yearold boy shows a well-defined, low-attenuation mass in the typical location of a Bailey type II 2nd branchial cleft cyst: Anterior to t…



Coronal CECT in the same patient demonstrates a thicker wall inferiorly, which histologically represented lymphoid follicles in the wall of the…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 602.

NECK: TRIANGLES, FASCIA & NODES · CONGENITAL

Lymphatic Malformation (Cystic Hygroma, Lymphangioma)

IMAGING FINDINGS

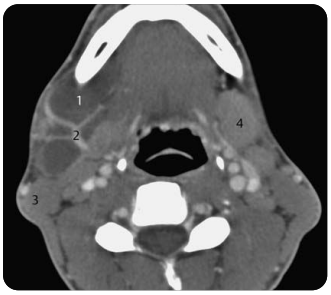
- Multicystic, septated, transspatial mass
- Fluid–fluid levels after bleeding
- Insinuates between structures without mass effect

TOP DIFFERENTIALS

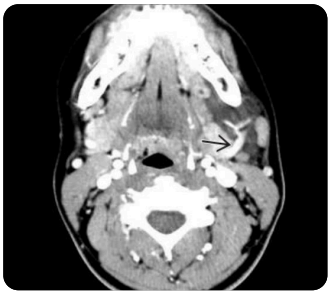
- Infantile haemangioma
- Venous malformation (phleboliths)
- Abscess

CHECKLIST

- Airway and mediastinal extent
- Macrocystic versus microcystic (sclerotherapy)



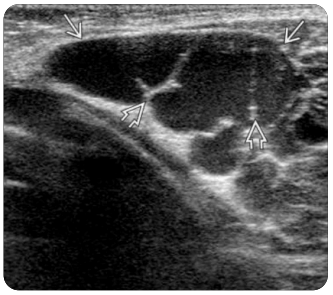
CT with intravenous contrast enhancement, axial. There is hypodense multilobular cystic structure (1) at the level of the…



Axial CECT in a 22-yearold woman with an angle of a mandible mass reveals a low-density submandibular space
Koch et al., Diagnostic Imaging: Head…



Axial T2 FS MR in the same patient shows the high fluid signal of the LM with its lobular margins. In this case, no fluidfluid levels are seen.…



Ultrasound in a 2-yearold patient with a soft, pliable mass shows a well-defined, lobulated, multicystic lesion with thin internal septations, typical o…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 580.

LARYNX & TRACHEA · TUMOURS

Supraglottic Tumor

IMAGING FINDINGS

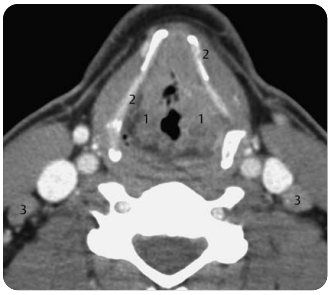
- Mass on the epiglottis, aryepiglottic folds or false cords
- Invasion of pre-epiglottic or paraglottic fat (T3)
- Nodes common, often bilateral

TOP DIFFERENTIALS

- Laryngocele
- Benign mucosal swelling
- Minor salivary tumour

CHECKLIST

- Vallecula and tongue base; pyriform sinus
- Cartilage and extralaryngeal spread



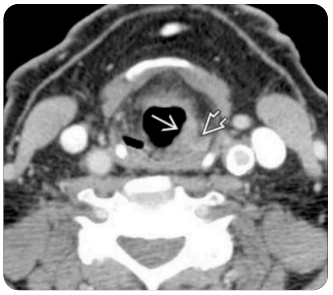
CT with intravenous contrast enhancement, axial. The lesion extends bilaterally (1), highly indicative of a malignancy. The…



Axial CECT shows similar findings with invasion of bilateral paraglottic fat and left AE fold. Tumor bulges through thyroid cartilage notch. Pyriform sinu…



Axial CECT shows small epiglottic mass, staged as T3 due to preepiglottic space invasion. Mass is symmetric, making it hard to…



Axial CECT shows small T1 posterior laryngeal SCCa with mass on left AE fold. Left pyriform sinus is collapsed, but no tumor is present in…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 526.

LARYNX & TRACHEA · TUMOURS

Glottic Tumor

IMAGING FINDINGS

- Thickening or mass of the true cord
- Anterior commissure thicker than 1 mm
- Paraglottic and subglottic spread

TOP DIFFERENTIALS

- Benign cord lesion
- Chondrosarcoma (cricoid, chondroid matrix)
- Cord paralysis (medial arytenoid)

CHECKLIST

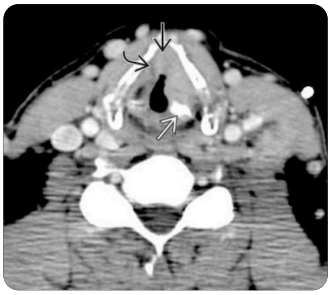
- Cord movement on breath-hold images
- Thyroid cartilage erosion: inner cortex T3, through it T4a



CT with intravenous contrast enhancement, axial. Slight asymmetry of the vocal cords can be observed with slight enlargement of the vocal…



Axial CECT shows an enhancing right TVC exophytic mass. Anterior and posterior commissures are normal. Right arytenoid cartilage sclerosis is…



Axial CECT reveals SCCa involving the entire left true TVC, anterior commissure, and anterior 1/3 of right cord. Left arytenoid and thyroid…



Axial CECT shows bulky, ulcerated anterior SCCa involving both vocal cords and anterior commissure. Both anterior thyroid cartilages are sclerot…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 530.

THYROID & PARATHYROID · CONGENITAL

Thyroglossal Duct Cyst

IMAGING FINDINGS

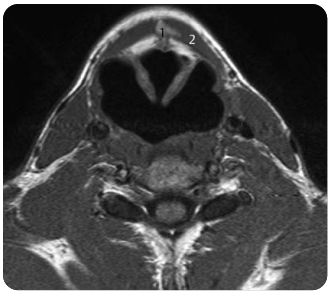
- Midline or paramedian cyst at or below the hyoid
- Infrahyoid cysts embedded in the strap muscles
- Thick wall or enhancement if infected

TOP DIFFERENTIALS

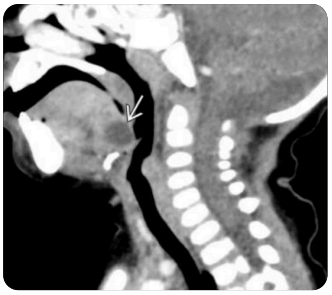
- Dermoid or epidermoid
- Lymphatic malformation
- Lingual thyroid

CHECKLIST

- Normal thyroid in the lower neck on ultrasound
- Solid nodule in the wall in an adult: carcinoma



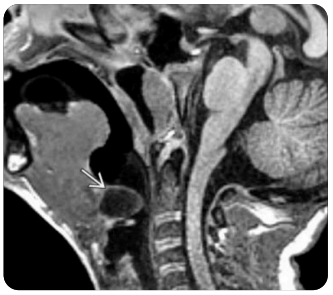
MRI, T1-weighted, axial. Small lesion (1) with high signal intensity on T1-weighted images, probably due to presence of protein-rich fluid, situate…



Sagittal CECT shows a well-defined cysticappearing mass at the midline base of the tongue. This was incidentally found on a CT performed…



Sagittal CECT in a child with sore throat and difficulty swallowing shows a lobulated low-attenuation mass in the midline sublingual space, with…



Sagittal T1WI in a child with recurrent TGDC demonstrates a cystic-appearing mass at the midline base of the tongue. Note distortion of the…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 590.

THYROID & PARATHYROID · INFECTION & INFLAMMATION

Hypertrophy of the Thyroid Gland, Thyroid Goiter

IMAGING FINDINGS

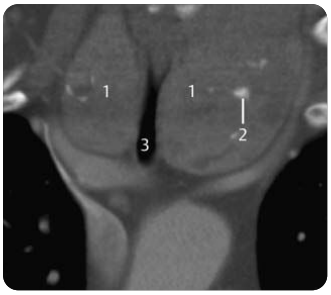
- Enlarged, heterogeneous gland with nodules, cysts and coarse calcification
- Tracheal deviation and narrowing
- Retrosternal extension

TOP DIFFERENTIALS

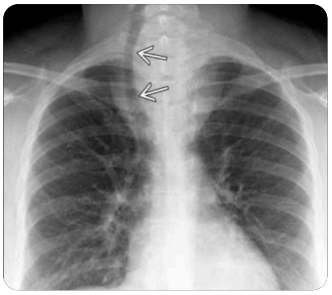
- Thyroid adenoma
- Differentiated or anaplastic carcinoma
- Lymphoma (rapid growth, Hashimoto)

CHECKLIST

- Lowest extent relative to the aortic arch and brachiocephalic vein
- Tracheal calibre for anaesthesia



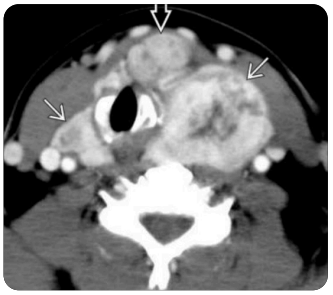
CT with intravenous contrast enhancement, coronal. This slice at the level of the lower neck/upper thoracic cavity, shows enlarged thyroi…



Posteroanterior chest radiograph demonstrates marked displacement of cervical and thoracic trachea by a large, left-sided neck & superior…



Coronal reformatted CECT in the same patient illustrates craniocaudad extent of the MNG with heterogeneous thyroid tissue bilaterally, but more…



Axial CECT at level of cricoid shows markedly heterogeneous, lobulated, and enlarged thyroid lobes as well as thyroid isthmus. No calcifications or…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 226.

LARYNX & TRACHEA · TRAUMA

Laryngeal Trauma

IMAGING FINDINGS

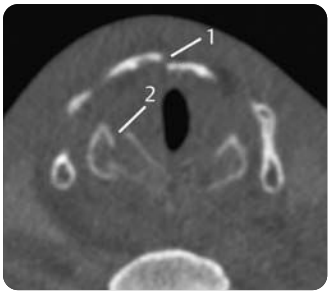
- Thyroid and cricoid cartilage fractures with displacement
- Arytenoid dislocation
- Haematoma and air narrowing the airway

TOP DIFFERENTIALS

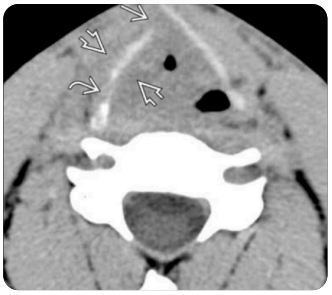
- Cord paralysis
- Unossified cartilage mimicking a fracture

CHECKLIST

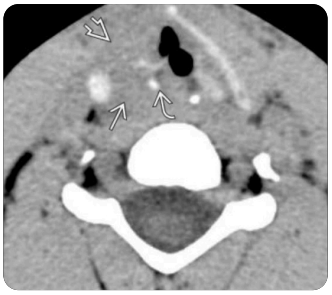
- Secure airway before scanning
- Cricoid fractures and displaced fragments usually need repair



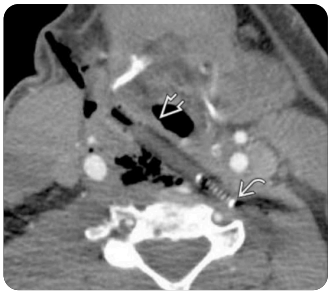
CT, axial bone window. CT at the level of the glottis showed multiple fractures; this figure shows a midline fracture with slight inward…



Axial NECT through the supraglottic larynx shows step-off at the superior thyroid notch. There is a subtle fracture of the right thyroid lamina…



Axial NECT more inferiorly in same patient, just above the cricoarytenoid joints, shows the anteromedial displacement of the right arytenoid…



Axial CTA performed for a pen stabbing through the supraglottic larynx shows displaced edematous right aryepiglottic fold. The pen tip…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 278.

LARYNX & TRACHEA · INFECTION & INFLAMMATION

Laryngocele

IMAGING FINDINGS

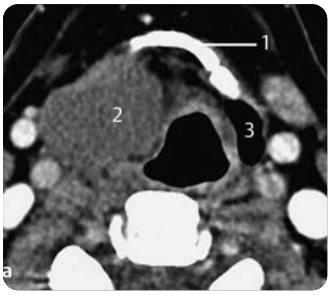
- Air or fluid-filled sac from the laryngeal ventricle
- Internal (paraglottic) or external through the thyrohyoid membrane
- Rim enhancement if infected (pyolaryngocele)

TOP DIFFERENTIALS

- Thyroglossal duct cyst
- Second branchial cleft cyst
- Saccular cyst

CHECKLIST

- Exclude a carcinoma at the ventricle in adults
- Airway compromise



CT with intravenous contrast enhancement, axial. The axial slice at the lower border of the hyoid (1) reveals a homogeneous, expanding…

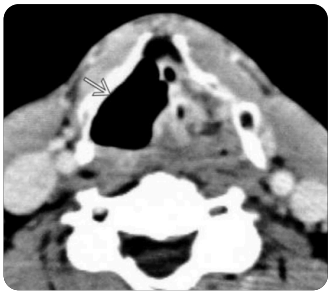


Coronal CECT reformat shows a similar lesion. This is also known as a "mixed" laryngocele because it contains internal…



Axial CECT demonstrates the typical findings of a fluid-filled, internal

Koch et al., Diagnostic Imaging: Head and Neck, 3rd ed., p. 292



Axial CECT in a professional trumpet player shows an airfilled sac confined to the right paraglottic region. Findings are typical of an air-filled, internal…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 292.

LARYNX & TRACHEA · INFECTION & INFLAMMATION

Paralysis of the Recurrent Laryngeal Nerve

IMAGING FINDINGS

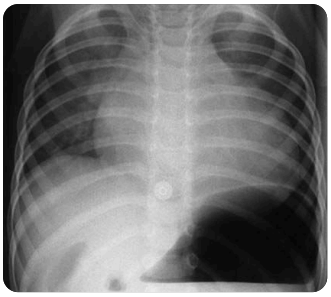
- Medially rotated arytenoid and dilated ventricle
- Dilated pyriform sinus; medial, thickened aryepiglottic fold
- Atrophy of the posterior cricoarytenoid and thyroarytenoid

TOP DIFFERENTIALS

- Arytenoid dislocation after trauma
- Glottic carcinoma
- Laryngocele

CHECKLIST

- Scan from skull base to the aortopulmonary window on the left
- Thyroid, oesophagus and mediastinal nodes



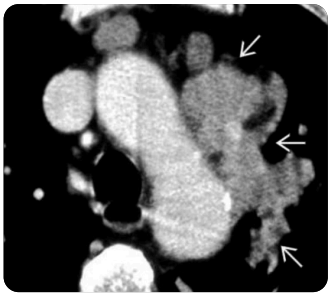
A 7-year-old child with hoarseness and left vocal cord paralysis. Chest radiograph, posteroanterior view. The grossly dilated heart is compressing…



CECT in a patient with new-onset hoarseness shows slightly medially rotated left arytenoid cartilage & a more prominent dilated left laryngeal…



Axial CECT in the same patient demonstrates asymmetrically enlarged left pyriform sinus and a medial position of the left aryepiglotti…



Axial CECT in same patient stresses the importance of scanning the aortopulmonary window for left VCP. A mediastinal mass…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 296.

THYROID & PARATHYROID · TUMOURS

Thyroid cancer

IMAGING FINDINGS

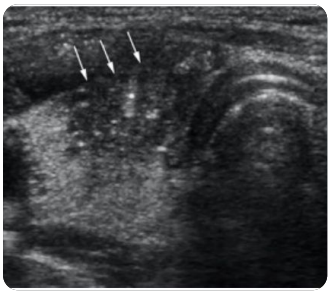
- Solid hypoechoic nodule with microcalcifications, taller than wide
- Irregular margins or extrathyroidal extension
- Cystic or calcified nodes in levels III, IV and VI

TOP DIFFERENTIALS

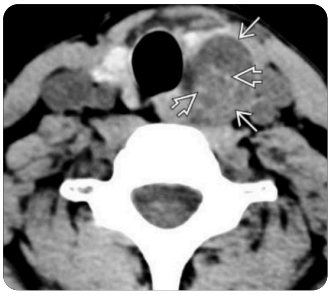
- Benign adenoma or colloid nodule
- Multinodular goitre
- Medullary or anaplastic carcinoma

CHECKLIST

- Trachea, oesophagus, larynx and recurrent nerve
- Iodinated contrast delays radioactive iodine



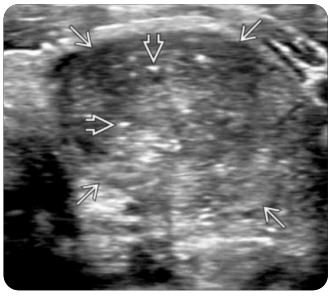
16-year-old lady presenting with hard right neck lump confirmed to be papillary ca on cytology. US shows strongly hypoechoic lesion with…



Axial NECT shows a well-defined mass arising from the left thyroid lobe with fine, speckled microcalcifications centrally. Small calcifications such as…



Axial T1 MR in a 49-yearold man with an enlarging neck mass shows a heterogeneous solid and cystic mass arising in left thyroid lobe, displacing…



Longitudinal thyroid ultrasound in a different patient reveals a well-defined mass with multiple tiny, hyperechoic microcalcifications, found to be a…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 238.

IMAGING

Advanced and rare

Once the core is secure. Each has its full set of images in the app.

Lymphadenopathy, HIV

Neck: triangles, fascia & nodes · Infection & inflammation

HIV infections may be associated with coexisting lymphoma and chronic lymphadenitis. Coexisting diseases may have a different clinical presentation in HIV.

Supraglottic and Pharyngeal Tumors

Larynx & trachea · Tumours

Staging is based on enlarged or pathologic-appearing lymph nodes, bilaterality, and overgrowth of the tumor on the contralateral side. · Lymphomas (non-Hodgkin) initially might present as unilateral lymphadenopathy and ulcerative lesions.

Aspiration

Larynx & trachea · Infection & inflammation

Occasionally aspiration is caused by mechanical failure of the epiglottis to close because of the presence of tumor, but most cases are due to neurologic (sensory) disorders without any visible pathology apart from the aspiration. Beware of aspiration pneumonia. · The above-mentioned pathologies are partly demonstrated in other sections.

Pneumothorax after Neck Surgery

Neck: triangles, fascia & nodes · Trauma

Most cases of pneumothorax are much smaller than in the picture shown here, and most of them are located at the lung apex.

Lipoma of the neck

Neck: triangles, fascia & nodes · Tumours

Schwannomas of the oral cavity and neck

Neck: triangles, fascia & nodes · After surgery

For personal study. Follow local protocols and senior advice for patients.